LIBBY — PATIENT/CAREGIVER TRACK

osteosarcoma-mets-dll3-h7r2

Plain-language summary for patient and caregiver

Generated 2026-08-21

This is decision-support information, not a treatment plan. Please share it with your oncologist before making any decisions based on what's here.

What this page is

You (or your clinician) asked Libby a narrow question. In metastatic osteosarcoma where there is a hint that both DLL3 and PRAME might be present, what treatments could go after either of those markers? You also said you would consider high-risk options when the possible payoff is real, and that lens shaped the whole report.

One thing comes before everything else. Both markers in your case were seen at the RNA level. RNA tells us a gene is being read inside the cell. It does not tell us the actual protein is sitting where these drugs need to find it. For the DLL3 drugs, the protein has to be on the outside of the cancer cells. For the PRAME drugs, the protein has to be present inside the cells and your immune system has to carry a specific tissue marker called HLA-A*02:01, because those drugs are built to work only when that marker is there. So the page has two parts. First, a small set of confirmation tests. Then treatment options that only open up depending on which tests come back positive.

The scope is deliberately tight. This page only covers drugs aimed at DLL3 or PRAME, because that was the question you asked. There are also standard, guideline-backed treatments for relapsed osteosarcoma that do not go after either marker. Those are not on this page. They are your oncologist's to walk you through, and they matter regardless of how these tests turn out.

What we know about your cancer

You are in your late teens or twenties with metastatic (stage IV) osteosarcoma, a cancer that started in bone. The plan assumes you have had standard frontline chemotherapy, the MAP regimen (methotrexate, doxorubicin, cisplatin), since that is the usual first treatment, but your inputs did not say how you responded or whether you have had anything since. Your day-to-day function reads as ECOG 1, which roughly means you are up and about but tire more easily than you used to. The two markers driving this report are the DLL3 and PRAME RNA signals, with the caveat from the section above about RNA not being the whole story.

What you told us matters most

You said you accept high-risk, high-reward options when the possible upside is meaningful. You did not name any side effect you wanted to rule out completely, and you flagged a clear preference for clinical trials. Those two things are why early-stage trial options, some of them first-in-disease, are on this page at all.

The first step everyone agreed on

Before any treatment decision, three tests need to happen. They can all run at once off the same stored biopsy block plus a single blood draw, so this is one order, not three separate trips.

1. **DLL3 protein test on the tumor** (an IHC stain using an antibody called SP347). This checks whether DLL3 is actually on the surface of the cancer cells, and at what level.
2. **PRAME protein test on the tumor** (a different IHC stain). This checks whether the PRAME protein is really being made.
3. **HLA-A*02:01 typing on a blood sample.** This checks whether your immune system carries the tissue marker that all the PRAME drugs are built around.

None of these tests is toxic. They usually run on tissue left over from a previous biopsy, so a fresh procedure often is not needed. Turnaround is about one to three weeks, and the cost is small next to a single cycle of treatment.

Here is why this step leads the whole page. The RNA signal opens no trial by itself. These tests are what turn it into something you can actually enroll in. The two pathways are independent, so you might have neither, one, or both.

A couple of honest odds worth knowing up front. The PRAME protein test comes back positive in roughly half of osteosarcoma cases, so that one is more likely than not to hold. But HLA-A*02:01 is carried by only about 40 to 50 percent of people, and it is genetic, so if you do not have it, the entire PRAME set is off the table no matter how strongly PRAME stains. The DLL3 protein result in bone cancer is genuinely unknown ahead of time.

If DLL3 comes back negative, the DLL3 options below close. If the PRAME test or the HLA typing comes back negative, the PRAME options close. And if all three are negative, this page has no further options to offer, because it was scoped only to these two markers. That is not the end of your care. Standard treatment for relapsed osteosarcoma is a real and separate conversation with your oncologist.

The options

There are six treatment options, plus the tests above. They split into two groups by which result has to be positive. Every one of them is a clinical trial, and every one is conditional on the confirmation tests. The DLL3 group comes first because DLL3 was the original signal in your file. The PRAME group is just as serious if both PRAME tests come back positive.

I want to be plain about the shape of the evidence here, because it runs through all six. These are immune therapies aimed at your markers, and the strongest data on any of them comes from other cancers, not osteosarcoma. Some are further along than others. None has published osteosarcoma results yet. That is the trade running through all six: real mechanism, real early signals, but a hypothesis when it comes to bone specifically.

Option 1 — Tarlatamab (DLL3 pathway)

This option carries caveats. It is only on the table if the DLL3 protein test comes back positive. If that test is negative or below the trial's threshold, this option is off.

When DLL3 is positive, this is the option to discuss first in the DLL3 group, and it rests on the strongest evidence of anything on this page.

Tarlatamab is a "bispecific T-cell engager." You get it as an infusion every two weeks. Think of it as a molecular clip: one end grabs DLL3 on the cancer cell, the other end pulls in one of your own T cells to attack it. It is already approved for small-cell lung cancer. The trial here is a "basket" study, meaning it takes any tumor that clears the DLL3 protein test, osteosarcoma included.

What might the upside look like? In small-cell lung cancer, where this drug is approved, a large randomized trial found that at the time of analysis roughly 40 percent fewer people had died on tarlatamab than on chemotherapy, and about 40 out of 100 patients had measurable tumor shrinkage. I cannot translate those lung-cancer numbers cleanly into an osteosarcoma expectation, because no one has published that data yet and it would be misleading to pretend I could. Whether the benefit carries over is exactly the question this trial exists to answer.

On the risk side, about half of patients get a reaction called cytokine release syndrome, a flu-like immune flare. Most cases are mild, and only about 1 in 100 are severe. Around 1 in 10 have neurologic side effects. The first cycle needs inpatient monitoring while the dose is stepped up. One point in its favor: this side-effect profile is well mapped, with a written plan for managing the immune flare, which matters more when the patient is young.

The open question is whether the lung-cancer results translate to bone at all. No osteosarcoma patient has been reported on this drug. Whoever enrolls is in the very first group. That is a real thing to weigh, not a footnote, and it is the trade against an otherwise strong track record.

Option 2 — Tarlatamab plus radiation (DLL3 pathway)

This option carries caveats. Same first lock as Option 1: the DLL3 protein test has to be positive. On top of that, this is an early trial with essentially no results yet, and it also needs the study to confirm bone sarcoma is eligible and that you have a tumor site suitable for radiation.

This is the same tarlatamab backbone from Option 1, with radiation added on. The idea is that radiating one tumor site might help the immune attack spread to others, an effect that has been seen occasionally in other settings. It is a reasonable hypothesis.

Here is the honest catch. There is no published data at all on giving tarlatamab and radiation together. Layering radiation onto a drug that can already cause immune flares and neurologic effects is exactly the kind of combined toxicity this trial was set up to measure, which means nobody can hand you a management plan for it yet. Option 1 gives you the same backbone with a worked-out safety plan behind it, which is why this combination sits lower. It is worth knowing about as a parallel DLL3 route, not as a first move.

Option 3 — IMA203 (PRAME pathway)

This option carries caveats. It is only available if both the PRAME protein test and the HLA-A*02:01 typing come back positive. Both have to be positive together. PRAME without the HLA marker, or the marker without PRAME, does not unlock it.

When both PRAME tests are positive, this is the option to discuss first in the PRAME group. It also carries the single strongest response number on this page, and the heaviest side-effect profile. Both of those are true at once, and that tension is the whole point of this option.

IMA203 is a "TCR-T cell therapy." Your own T cells are collected, taken to a lab, given a custom receptor that recognizes the PRAME target sitting on the HLA-A*02:01 marker, grown in large numbers, and infused back into you. The trial is a basket study that accepts any PRAME-positive, HLA-A*\02:01-positive solid tumor and names a sarcoma group specifically, so osteosarcoma fits in principle rather than by argument.

What might the upside look like? In the published early study, about 54 out of 100 patients across a mix of solid tumors had measurable tumor shrinkage, and in those who responded the benefit was still going at nine months. That is the biggest response signal in this whole report. Two honest limits, though. It comes from only 28 patients, which is a small number, so the true rate could be quite a bit higher or lower than 54 percent. And the patients who responded leaned toward other cancers, not bone. No osteosarcoma-specific response has been reported.

On the risk side, this is the most demanding option here. Nearly everyone gets cytokine release syndrome, and about 1 in 10 of those cases is severe. Around 1 in 4 have neurologic side effects. Before the cells go in, you get a short course of two chemotherapy drugs to make room for them, which drops your blood counts for several weeks and opens a real window of vulnerability to infection. In the published group of 28, one patient died from an infection during that window. That is not a number to rank past quietly. The custom cells also take about four to six weeks to make, so this is not a fast start.

The trade is stark. The biggest possible benefit on the page, paired with the biggest demands and the only reported death. If both markers confirm, a real question to work through with your team is whether the treating center has the cell-collection setup and the infection and immune-flare monitoring to do this safely.

Option 4 — Brenetafusp (PRAME pathway)

This option carries caveats. Like Option 3, it is only on the table if both the PRAME protein test and HLA-A*02:01 typing come back positive. On top of that, the osteosarcoma-relevant slot for this drug is not open right now, so reaching it depends on that group reopening or on the newer version in Option 5.

Brenetafusp goes after the same PRAME target on the same HLA-A*02:01 marker as IMA203, but it does it with a small off-the-shelf molecule called an "ImmTAC" rather than your own engineered cells. That makes it much simpler to give: a weekly infusion, no cell collection, no bone-marrow-clearing chemotherapy first. It is the best-tolerated route onto the PRAME pathway.

The gentleness comes with a cost in likely benefit. In melanoma, where most of the data comes from, about 9 to 11 out of 100 patients had measurable shrinkage. There was a hint that PRAME-positive patients did better on a couple of survival measures, but that hint comes from an unadjusted slice of an early conference report, not a finished peer-reviewed result, so it should not be read as a survival promise across different cancers. On side effects, most of the immune flares were mild with none severe in the melanoma group, though a rash is common and can be managed with a steroid premedication.

The open question is whether that modest melanoma benefit shows up in bone at all, and whether an eligible slot exists. The appeal is real: the easiest PRAME route to tolerate and deliver, off the shelf, no manufacturing wait. The catch is thin single-agent numbers and a currently closed door.

Option 5 — IMC-P115C (PRAME pathway)

This option carries caveats. Again, both the PRAME protein test and HLA-A*02:01 typing have to be positive. And because this is a very early trial, the study team has to confirm osteosarcoma is eligible before it is a real slot.

IMC-P115C is the next-generation version of the Option 4 drug, from the same maker and built on the same approach. It works the same way, a small off-the-shelf ImmTAC given as a weekly infusion, and its broader drug class is validated by an approved drug for a rare eye cancer.

Here is the honest state of it. There are no published results for this specific drug yet, because it is in its earliest human testing. So the efficacy expectation is borrowed from Option 4, and the borrowed survival hints there are early and unadjusted, which means they should not be treated as a promise for this drug either. What earns it a place is timing: it is actively recruiting, where Option 4's osteosarcoma slot is currently closed. If both PRAME tests confirm and you want the gentler PRAME route with an open door today, this is the one to ask about. The first step would be contacting the sponsor to confirm bone sarcoma is eligible.

Option 6 — NW-101C (PRAME pathway)

This option carries caveats. Same double lock as the other PRAME options: both PRAME protein and HLA-A*02:01 have to be positive. And it needs the sponsor to confirm osteosarcoma is eligible.

NW-101C is a second cell-therapy option on the PRAME pathway, the same general approach as IMA203 in Option 3. It sits at the bottom of the page for one plain reason: there is no published data of any kind under this drug's name yet. Searching for it turns up no clinical results. Everything we would expect from it is simply borrowed from IMA203.

So why is it here at all? Redundancy on your own marker. If the IMA203 slot in Option 3 is full or too far to reach, a second PRAME cell therapy is a genuine fallback for someone willing to travel. Treat it as a slot to verify with the sponsor, not as a confident pick. Expect the same demanding side-effect and blood-count profile as Option 3, since it uses the same kind of treatment.

Questions to ask your oncologist

1. Can we order all three confirmation tests at once, the DLL3 protein stain (SP347), the PRAME protein stain, and HLA-A*02:01 typing? They are the precondition for everything else, and they can run on one stored biopsy block plus one blood draw.
2. Once the results are in, can we walk through what each combination means? "DLL3 positive only," "PRAME and HLA both positive," "all three positive," and "all negative" are genuinely different conversations.
3. If all three come back negative, what standard options do you usually consider for relapsed osteosarcoma? This page was scoped only to DLL3 and PRAME, so it does not cover those, but they are real and I would want to understand them ahead of time.
4. If DLL3 is positive, what is the realistic path to the tarlatamab trial from here? I would want to know the nearest enrolling site, the usual wait, and what the inpatient first-cycle monitoring actually looks like.
5. If PRAME and HLA are both positive, how would you weigh the high-response but demanding cell therapy (Option 3) against the gentler off-the-shelf infusion (Options 4 and 5)? I want to understand that trade-off in my own situation.
6. For the cell therapy specifically, does the nearest center have the cell-collection setup and the infection-monitoring protocols to run it safely, and how long is the manufacturing wait there?
7. What is your hospital's plan for cytokine release syndrome and the neurologic reactions? Several of these options carry that risk, and how it is handled locally matters more than the published rates.
8. For the trials that need sponsor sign-off on osteosarcoma eligibility (Options 2, 5, and 6), would your team be

willing to contact the sponsors to confirm whether bone sarcoma is in scope?

Sources

The recommendations drew on the following published reports and clinical-trial records.

- Mountzios et al. *N Engl J Med* 2025. Tarlatamab DeLLphi-304 phase 3 in small-cell lung cancer (PMID 40454646).
- Ahn et al. *N Engl J Med* 2023. Tarlatamab DeLLphi-301 in small-cell lung cancer (PMID 37861218).
- Wermke et al. *Nat Med* 2025. IMA203 PRAME TCR-T pan-solid-tumor cohort (PMID 40205198).
- Yao et al. *Oncologist* 2022. DLL3 as a therapeutic target, review (PMID 35983951).
- Hamid et al. ASCO 2024, abstract 9507. Brenetafusp PRAME ImmTAC in melanoma (DOI 10.1200/JCO.2024.42.16_suppl.9507).
- Trial registry NCT06788938 (tarlatamab DLL3-IHC basket).
- Trial registry NCT06814496 (tarlatamab plus radiation, DLL3-IHC basket).
- Trial registry NCT03686124 (Immatics ACTengine IMA203 PRAME TCR-T basket).
- Trial registry NCT04262466 (brenetafusp PRAME ImmTAC).
- Trial registry NCT07156136 (Immunocore IMC-P115C PRAME ImmTAC).
- Trial registry NCT07266298 (NW-101C PRAME TCR-T).